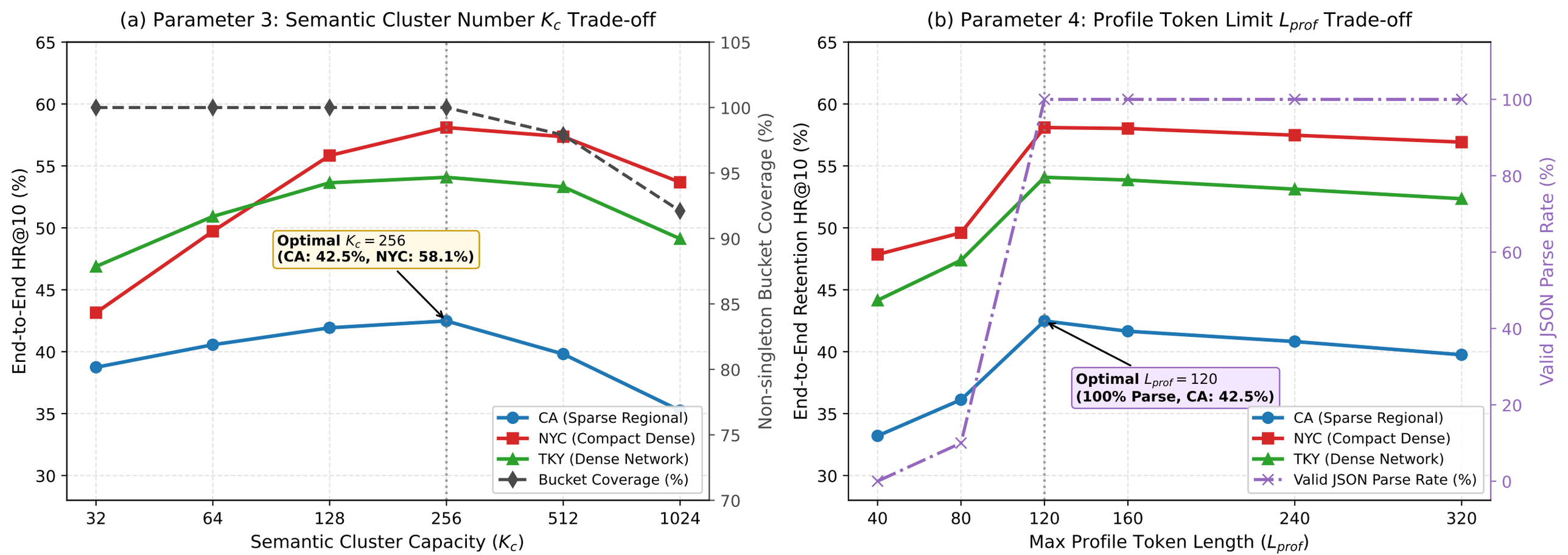
(a) Parameter 3: Semantic Cluster Number K_c Trade-off
End-to-End HR@10 (%)
Non-singleton Bucket Coverage (%)
Semantic Cluster Capacity (K_c)
Optimal $K_c = 256$
(CA: 42.5%, NYC: 58.1%)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Bucket Coverage (%)
(b) Parameter 4: Profile Token Limit L_{prof} Trade-off
End-to-End Retention HR@10 (%)
Valid JSON Parse Rate (%)
Max Profile Token Length (L_{prof})
Optimal $L_{prof} = 120$
(100% Parse, CA: 42.5%)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Valid JSON Parse Rate (%)